CSE 3422: PROJECT REQUIREMENTS

All teams must develop and deploy a full-stack software application while following proper software engineering practices throughout the project.

1. **Frontend and Backend Development:** The project must use a frontend framework/library and a backend framework of the team's choice. The application must include a functional user interface and a backend that handles the application's business logic.

2. **Team Formation:** Each team must consist of a minimum of 2 members and a maximum of 5 members. Each member must contribute meaningfully to the project.

3. **Version Control with Git:** Teams must use **Git** for version control and a shared remote repository (e.g., **GitHub or GitLab**). The repository should demonstrate regular commits, meaningful commit messages, and appropriate use of branches and merges.

4. **Project Management with Jira:** Teams must use **Jira** to manage project activities. The Jira board should include tasks or user stories, task assignments, and progress tracking throughout development.

5. **API Integration:** The application must include at least one meaningful feature that integrates an external API. Examples include:
    - Maps or location services
    - Generative AI services, such as **Gemini** or **OpenAI**
    - Weather or other public data APIs
    - Payment gateways, such as **SSLCommerz** or **Stripe**, using test/sandbox mode where applicable

Teams may choose other suitable APIs. Any free-tier or usage restrictions must be considered.

6. **API Testing:** Teams must use an API testing tool, such as **Postman** or **Insomnia**, to test the APIs used in their project. They should be able to demonstrate relevant requests, responses, and test results.

7. **Deployment:** The completed application must be deployed using a free hosting service or an available free tier such as **Vercel** or **Render**. Both the frontend and backend must be accessible in the deployed application, as applicable. If the chosen service has limitations, teams must ensure their application remains demonstrable.

8**. Application Testing:** Teams must use at least one testing tool or framework of their choice to test their application. Examples include **Selenium** for browser-based testing, unit-testing frameworks, integration-testing tools, or other suitable testing tools. Teams must demonstrate their tests and provide evidence of the testing performed, including the test cases and their results.

9. **Individual Contribution:** Every team member must be able to explain the parts of the project they contributed to and demonstrate an understanding of the overall system. Individual evaluation will be conducted during the demonstration.